AI-POWERED CUSTOMER CHURN PREDICTION

Data Analytics & Artificial Intelligence Internship – Final Project

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Executive Summary

This project demonstrates an end-to-end data analytics and AI workflow for a subscription-based business. The objective is to understand customer churn, identify the factors associated with churn, predict which customers are most likely to leave, and convert those predictions into actionable retention strategies. The workflow covers data preparation, exploratory analysis, KPI reporting, machine learning, risk scoring, and business recommendations.

Business Problem

Customer churn directly reduces recurring revenue and increases the cost of acquiring replacement customers. A business needs an early-warning system that prioritizes customers who show a high probability of churn so that retention teams can intervene before the customer leaves.

Project Objectives

• Measure overall churn and compare churn across plans, contracts and regions.

• Identify behavioral and service-related variables associated with churn.

• Build predictive models to estimate individual customer churn probability.

• Create practical risk bands: Low, Medium and High.

• Estimate revenue exposure and recommend targeted retention actions.

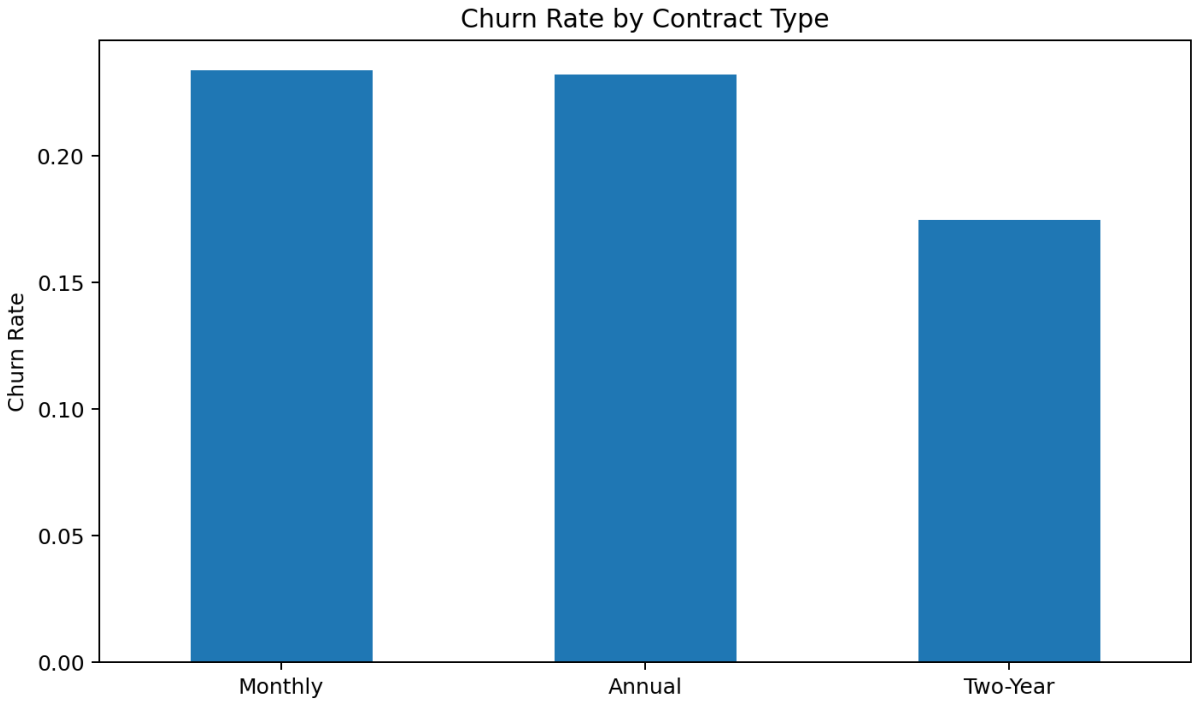
Dataset & Data Preparation

The project contains 5,000 customer records and 17 source fields covering demographics, subscription details, usage, billing, service interactions and satisfaction. The dataset used for this portfolio project is synthetic, which avoids exposing real customer information while preserving realistic business relationships.

Data preparation included type validation, categorical encoding, numerical scaling, missing-value handling through pipeline-based imputation, and a stratified 80/20 train-test split.

Key KPIs

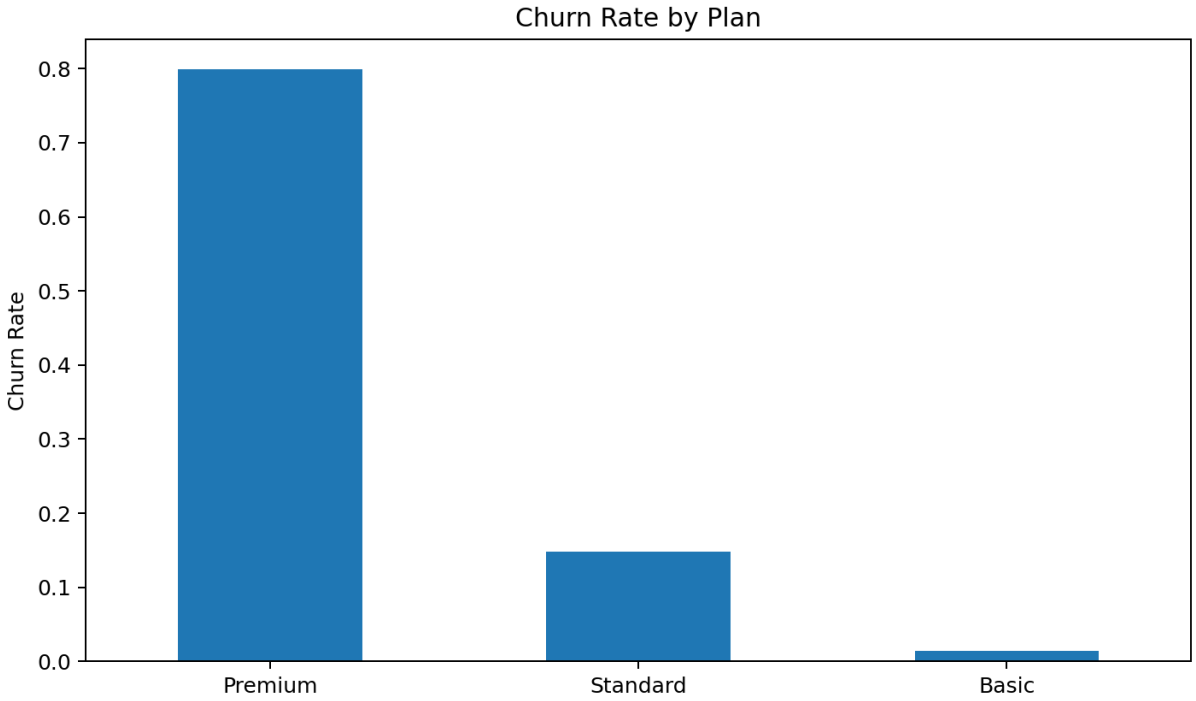
Metric	Result
Total Customers	5,000
Churned Customers	1,121
Overall Churn Rate	22.4%
Average Monthly Fee	INR 724
Average Satisfaction	3.48/5
High-Risk Customers	880
6-Month Revenue at Risk	INR 6,134,220


Churn Rate by Contract Type
Churn Rate
0.00
0.05
0.10
0.15
0.20
Monthly
Annual
Two-Year

Exploratory Data Analysis Findings

The analysis indicates that contract structure is a major business-level separator of churn. Monthly customers show higher churn exposure than customers on longer contracts. This suggests that retention programs should focus first on customers with flexible monthly commitments, especially when combined with low satisfaction or repeated support interactions.

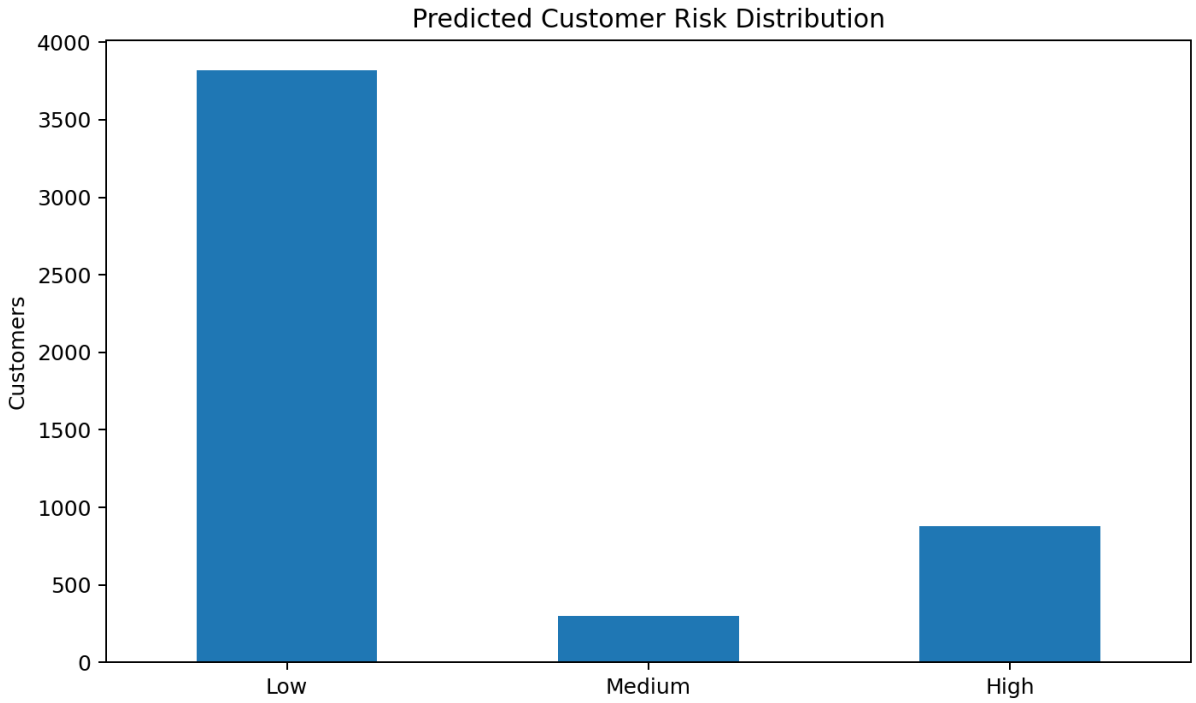
Plan-level analysis provides a second segmentation layer. The business should not treat all customers equally: high-risk customers with meaningful recurring revenue deserve more intervention than low-value low-risk accounts.



AI / Machine Learning Approach

Two classification models were evaluated: Logistic Regression as an interpretable baseline and Random Forest as a non-linear ensemble model. The final model was selected using ROC-AUC on the held-out test set. This avoids choosing a model solely because of raw accuracy, which can be misleading when churn classes are imbalanced.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.885	0.791	0.661	0.720	0.913
Random Forest	0.871	0.719	0.696	0.707	0.900


Predicted Customer Risk Distribution
Customers
4000
3500
3000
2500
2000
1500
1000
500
0
Low
Medium
High

Business Recommendations

1. Prioritize monthly-contract customers. Offer a structured upgrade path, loyalty benefit or time-bound retention incentive before renewal risk peaks.

2. Use satisfaction as an intervention trigger. Customers with low satisfaction should enter a service-recovery workflow before a sales discount is offered.

3. Reduce repeated support friction. High support-ticket customers should be reviewed for unresolved issues, onboarding gaps or product usability problems.

4. Combine probability with value. The highest priority should be customers who are both high-risk and commercially important. This creates a risk-value matrix instead of treating every churn prediction equally.

5. Close the analytics loop. Track retention campaign response, churn prevented, revenue saved and false-positive interventions so the model can be recalibrated over time.

Suggested Business Workflow

Data → Cleaning → EDA → KPI Dashboard → Churn Model → Risk Score → Customer Segmentation → Retention Action → Outcome Tracking

Limitations & Next Steps

Because the dataset is synthetic, the model metrics should be interpreted as a demonstration of methodology rather than evidence of production performance. A production implementation should retrain on verified historical customer data, perform cross-validation and calibration, monitor drift, and evaluate the financial impact of interventions through controlled experiments.

Conclusion

The project demonstrates how data analytics and AI can move from descriptive reporting to predictive decision support. Instead of only asking which customers have already churned, the proposed system identifies customers who may churn next, estimates potential revenue exposure, and provides a prioritization framework for retention teams.